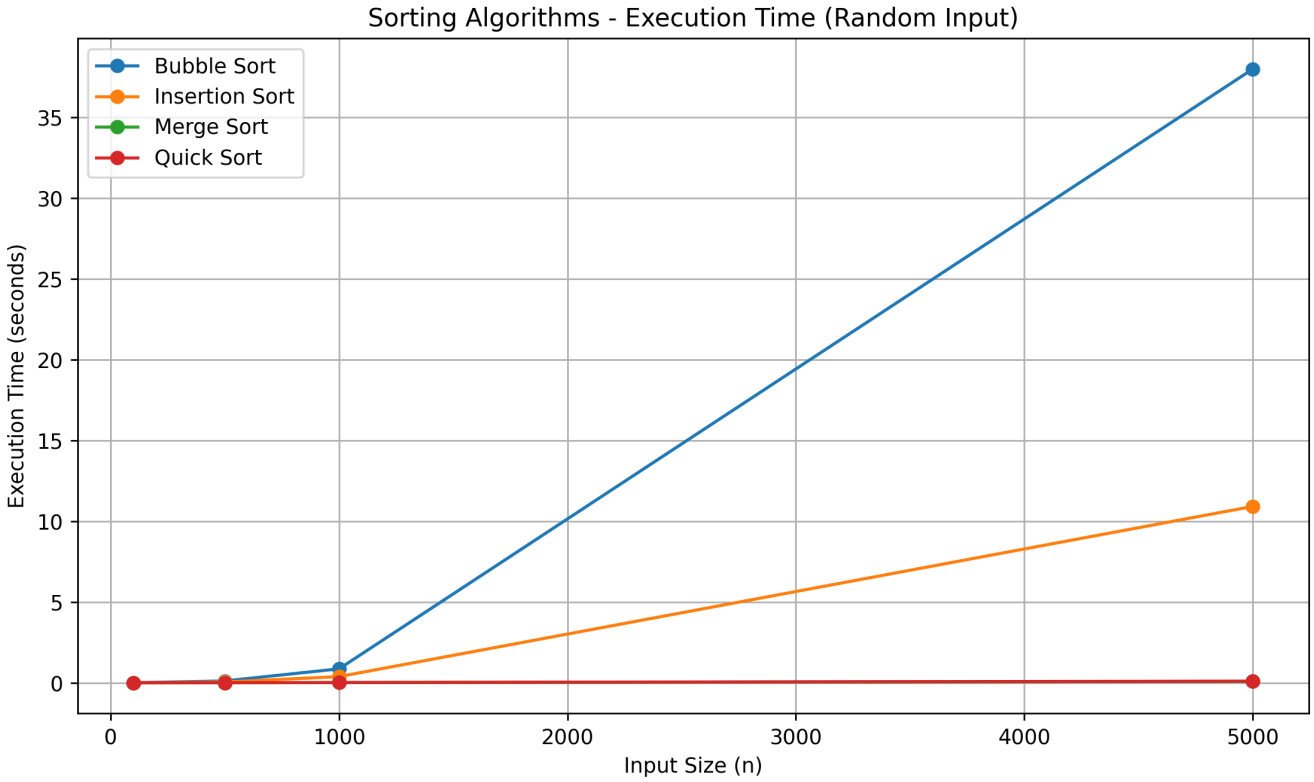
DAA Lab 1
Graphs and Visualizations

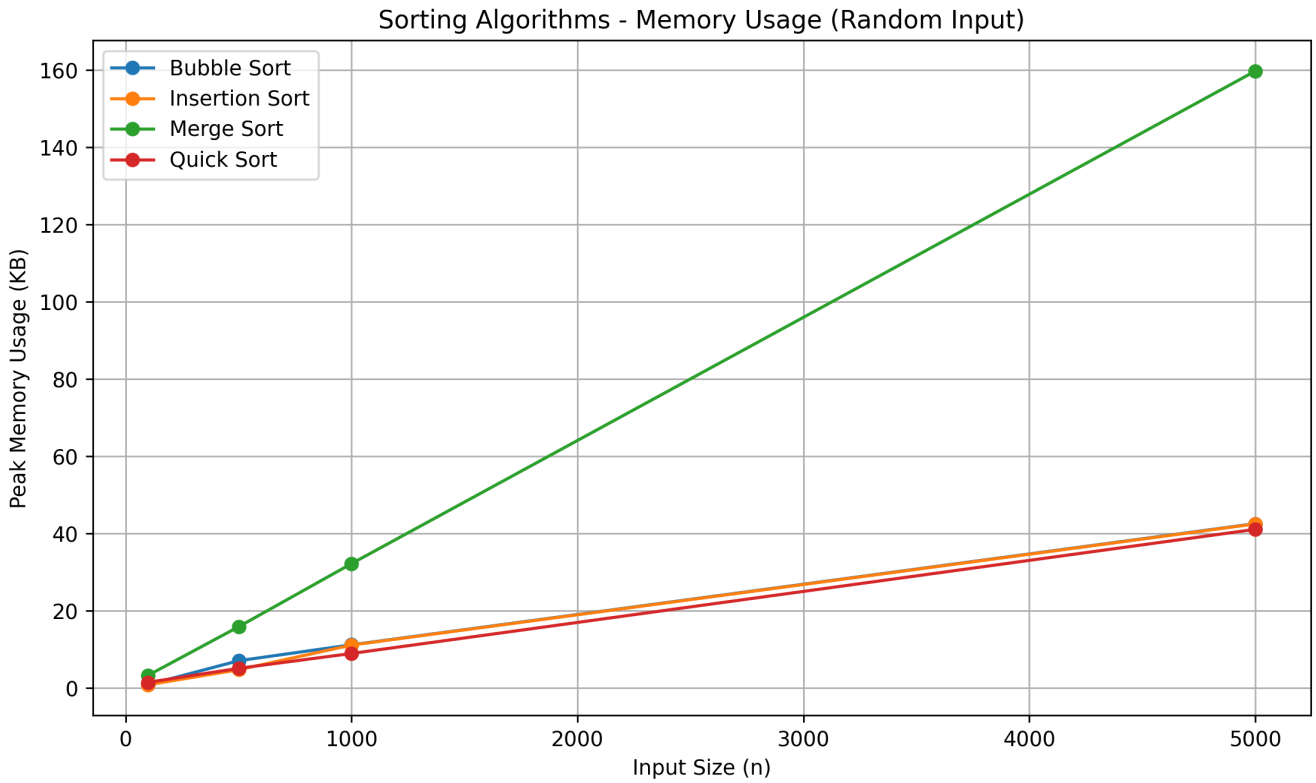
The sorting graphs use the complete random-input measurements through n=5,000. The 10,000 sorting run is incomplete, so it is intentionally excluded from the plotted comparison rather than assigning estimated values.

1. Sorting – Execution Time



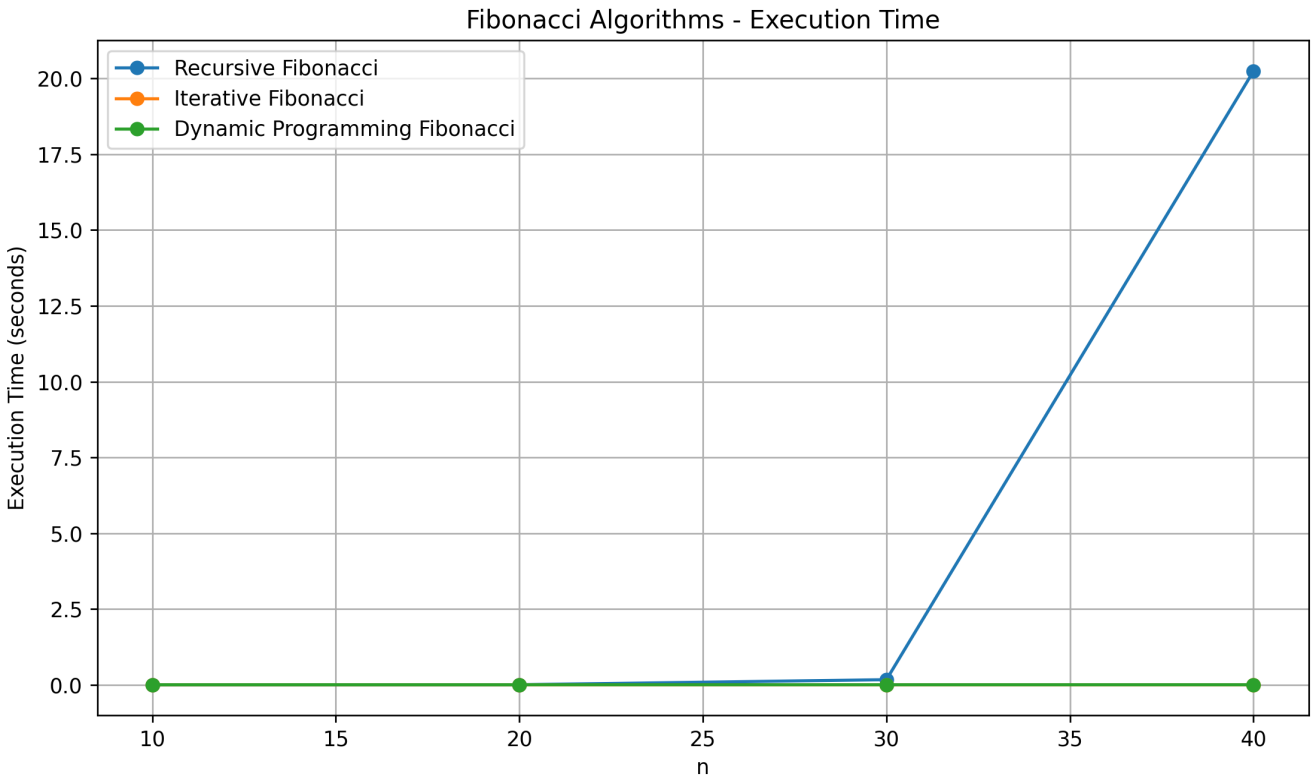
Observation: Merge Sort remains relatively stable as n grows. Bubble and Insertion Sort rise much more sharply at 5,000 random elements. Quick Sort is also fast on random data in this experiment.

2. Sorting – Memory Usage



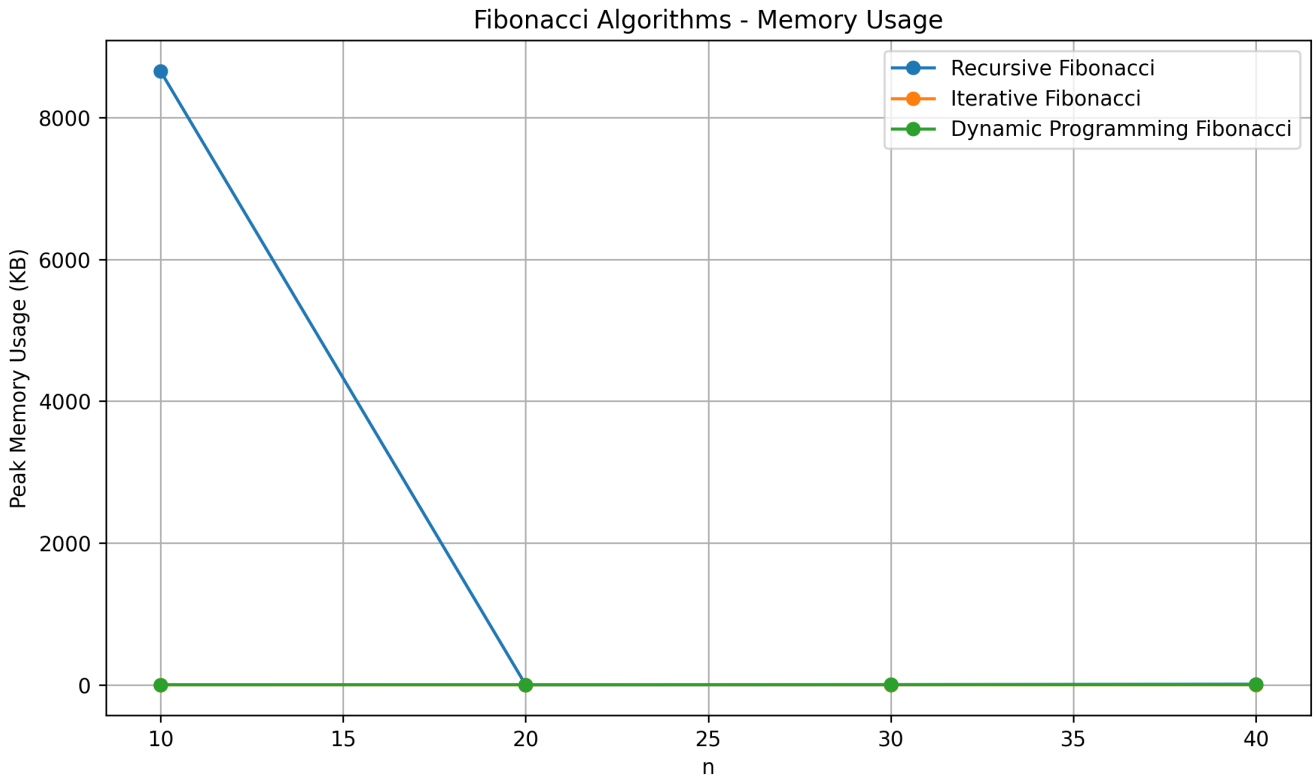
Observation: Merge Sort uses more auxiliary memory than the in-place Bubble/Insertion/Quick implementations. The measured values also include Python runtime and tracemalloc effects.

3. Fibonacci – Execution Time



Observation: Recursive Fibonacci grows exponentially in practice and reaches 20.2445 seconds at n=40. Iterative and DP implementations remain extremely fast for the tested range.

4. Fibonacci – Memory Usage



Observation: Iterative Fibonacci has the lowest and most stable measured memory usage. DP uses additional memory to store previously computed values. The recursive n=10 measurement is unusually high compared with later recursive points and should be treated as a measurement artifact/overhead rather than a clean representation of asymptotic space growth.